

CSE-420

Final Exam

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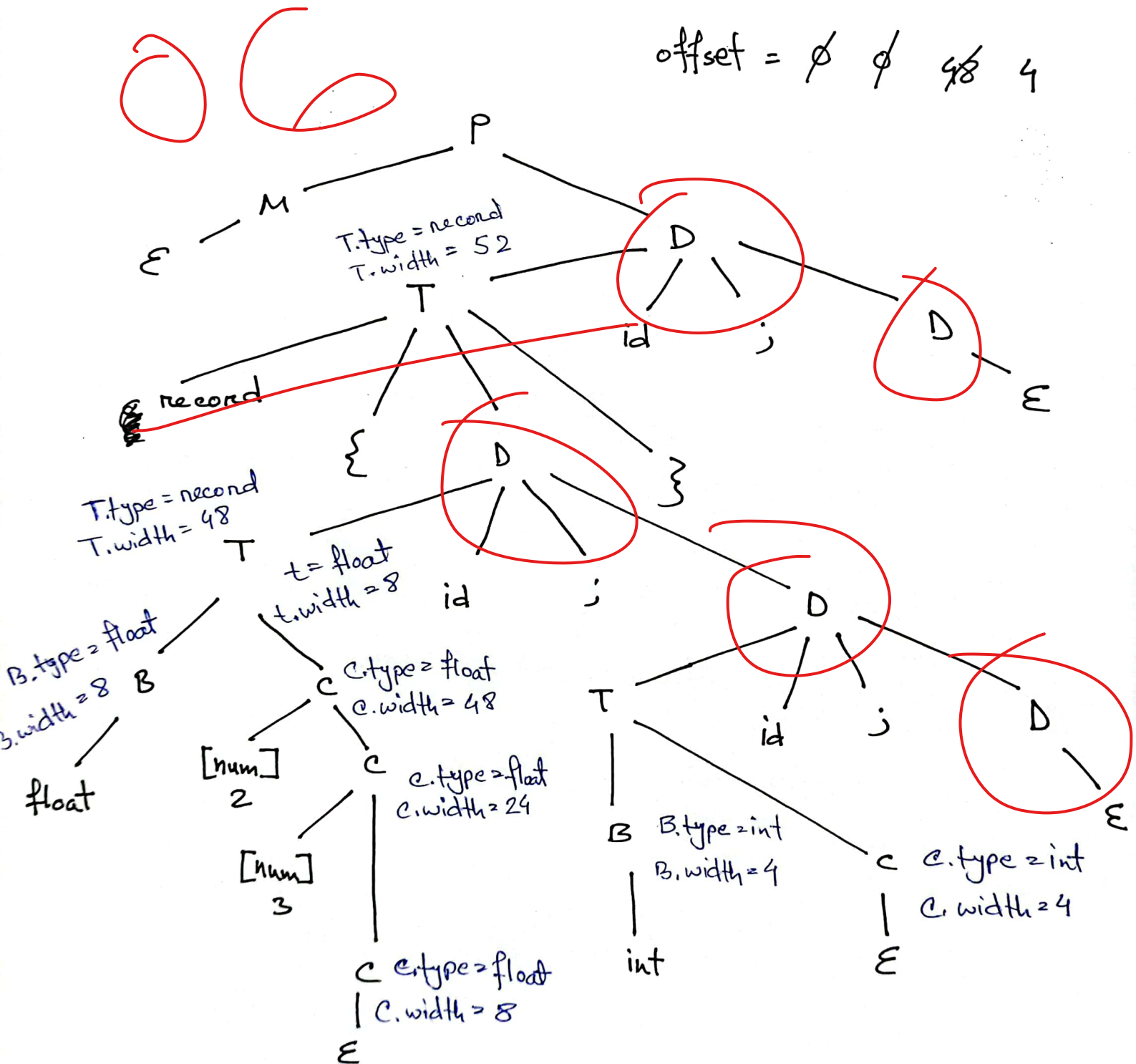
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Ans. to the question no. 1

record { float [2][3] a; int b; } p;

offset = 0 0 48 4



Ans. to the question no. 2

Given, CPU is executing the line ' $r = r * m$ '.
We are to draw the contents of active symbol tables at that point.

Here,

Global Scope :

- i. x : float (0.0)
- ii. v : float (0.0)
- iii. m : int [10][20]

Main function Scope :

- i. argCount : int (value depends on program arguments)
- ii. argValues : char*[] (value depends on program arguments)
- iii. t : float (1.0)
- iv. r : int [10] (updated while loop)

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Inner 'for' loop Scope :

- i. $i : \text{int} (\text{current loop iteration value})$
- ii. $s : \text{float} (0.0)$
- iii. $c : \text{int} (20)$
- iv. $m : \text{int} (\text{value of } m[i][j])$
(reused variable name, different scope)
- v. $\text{count} : \text{int} (0)$

Inner 'while' loop Scope :

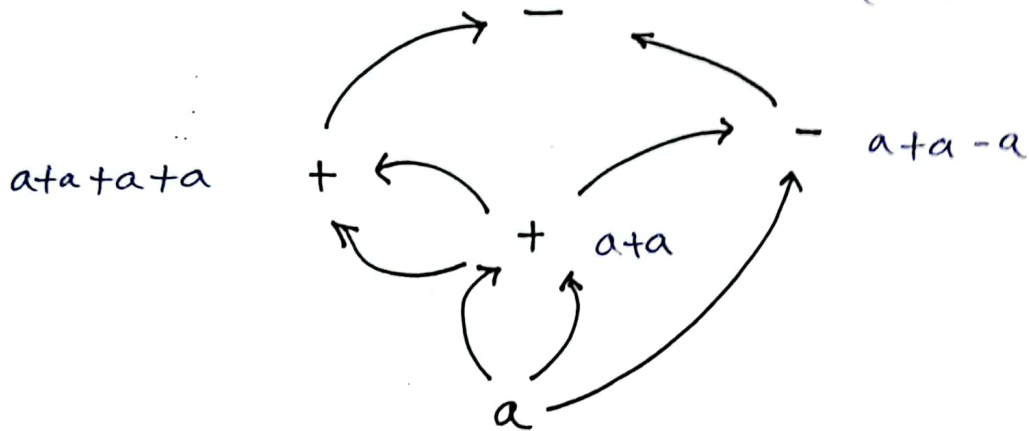
- i. $r : \text{float} (1.0)$ (reused variable name, different scope).
- ii. $\text{count} : \text{int}$
- iii. ~~$m : \text{float}$~~ (Shadows global 'm' and inner loop 'c')

Now drawing a single list of hash-tables:

Global scope → Main function scope → Inner for loop scope →
Inner while loop scope

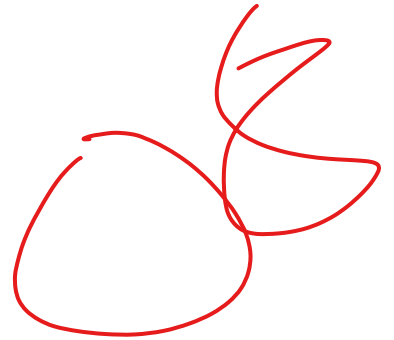
Ans. to the question no. 3

$$(a+a+a+a) - (a+a-a)$$



$$(a+a+a+a) - (a+a-a)$$

1. variable $\rightarrow a$
2. temporary register
3. constant $\rightarrow$ no constant

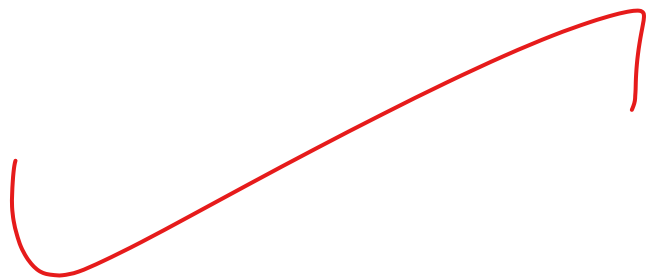


$$\Rightarrow t_1 = a + a$$

$$t_2 = t_2 + t_1$$

$$t_3 = t_1 - a$$

$$t_4 = t_2 - t_3$$



Quadruples :

op	Arg 1	Arg 2	result
+	a	a	$t_1 = a + a$
+	t_1	t_1	$t_2 = t_1 + t_1$
-	t_1	a	$t_3 = t_1 - a$
-	t_2	t_3	$t_4 = t_2 - t_3$

Triples :

op	Arg 1	Arg 2
(0) +	a	a
(1) +	(0)	(0)
(2) -	(0)	a
(3) -	(1)	(2)